FACULTY OF COMPUTING

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SECJ 3553 - ARTIFICIAL INTELLIGENCE

SECTION 02

PROJECT:

SMART AGRICULTURE SYSTEM
(ASSIGNMENT 1)

THEME:

AGRICULTURE

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Table of Contents

Knowledge Representation

1. **Automation of Irrigation**

   An irrigation system to regulate water usage efficiently, reduce waste, and ensure crops receive the optimal amount of water based on soil condition and weather patterns.

   - *IF soil_moisture_level = low AND rain_prediction = false*
     THEN irrigation_status = activate
   - *IF soil_moisture_level = adequate OR rain_prediction = true*
     THEN irrigation_status = deactivate

   Explanation:

   This irrigation system uses sensors and weather data to determine whether crops have adequate moisture for proper growth. When the soil moisture is low and no rain is forecasted, the irrigation system activates and waters the plant. When the soil moisture is at a recommended level or rain is forecasted, the irrigation system does not activate to avoid overwatering the plants.

   Predicate Logic / First-Order Logic:
   Activate Irrigation

   $\forall x \forall y(\text{LowMoisture}(x) \land \neg\text{RainExpected}(y) \rightarrow \text{ActivateIrrigation}(x))$

   Deactivate Irrigation

   $\forall x \forall y(\text{AdequateMoisture}(x) \lor \text{RainExpected}(t) \rightarrow \text{DeactivateIrrigation}(x))$

   The first predicate is for activating the irrigation system, where irrigation is activated only when the soil moisture is low and no rain is expected. The second predicate deactivates the irrigation system when the soil has enough moisture or when it is expected to rain.

2. Electric and Autonomous Machinery

Using electric-powered and AI-driven autonomous machines to reduce emissions, improve efficiency, and increase farm safety.

- *IF machinery_battery_level = low OR terrain_condition = unstable OR sensor_health_status = faulty THEN autonomous_machine_operation = unsafe*
- *IF obstacle_distance = close AND braking_system_response = slow THEN collision_risk = high*

Explanation:

The system utilizes sensors to assess the physical state of a machine in order to maintain safe operations. Low battery, unstable terrain, or faulty sensors mean machines might not be functioning right, so the system calls it unsafe to operate and then calls for protective actions. The AI also scans the environment for obstacles. If an object is too close and the braking response is slow, it detects a high collision risk and slows down or stops the machine. These rules help prevent accidents and ensure reliable autonomous operation.

Predicate Logic / First-Order Logic:

1. $\forall x\ (\ \text{lowBattery}(x) \lor \text{unstableTerrain}(x) \lor \text{faultySensor}(x) \rightarrow \text{unsafeOperation}(x)\ )$
2. $\forall x\ \forall y\ (\ \text{closeObstacle}(x, y) \land \text{slowBraking}(x) \rightarrow \text{highCollisionRisk}(x, y)\ )$

The first predicate shows that any machine with a low battery, unstable terrain, or faulty sensors is automatically considered unsafe. The second predicate states that if an obstacle is close and braking is slow, the machine is at high risk of collision.

3. Predictive Crop Modeling

This goal emphasizes anticipating crop growth and future yield based on current and historical environmental patterns.

- *IF weather_conditions = extreme OR climate_pattern = unstable THEN yield_prediction = uncertain*
- *IF soil_status = poor OR historical_growth_trend = unstable THEN yield_prediction = low*
- *IF weather_conditions = favourable AND soil_status = good AND historical_growth_trend = stable THEN yield_prediction = high*

Explanation:

When weather conditions are extreme or when climate patterns become unstable, the system most likely produces uncertain yield predictions, which makes planning and resource allocation more challenging. Then, poor soil health or inconsistent past growth trends can also lead to lower yield forecasts. The system can analyze patterns from weather data, soil readings, and historical growth records to automatically predict the outcomes on crop yield.

Predicate Logic/First-Order Logic:

1. $\forall x\ ((\text{ExtremeWeather}(x) \lor \text{UnstableClimate}(x)) \rightarrow \text{Predict}(x, \text{UncertainYield}))$

For every crop instance, if the weather conditions are extreme or the climate pattern is unstable, then the system will predict that the crop yield is uncertain.

2. $\forall x\ ((\text{PoorSoil}(x) \lor \text{DecliningHistory}(x)) \rightarrow \text{Predict}(x, \text{LowYield}))$

For every crop instance, if the soil quality is poor or the historical growth trend is declining, then the system will predict a low crop yield.

3. $\forall x\ ((\text{FavorableWeather}(x) \land \text{GoodSoil}(x) \land \text{StableHistory}(x)) \rightarrow \text{Predict}(x, \text{HighYield}))$

For every crop instance, if the weather is favorable, soil quality is good, and the historical growth trend is stable, then the system will predict a high crop yield.

4. **Pest and Disease Prediction**

   Using AI-enabled sensing and crop-health monitoring to detect early signs of pest and disease threats to reduce pesticide usage and protect crop yield.

   - *IF leaf_spot = yes THEN apply_pesticide*
   - *IF humidity = high AND temperature = high THEN fungal_removal*

   Explanation:

   This system uses sensor data to detect early pest and disease risks in the system. If leaf spots are found, pesticide treatment is applied. When both humidity and temperature are high, fungal growth is likely, so fungal removal is performed. This helps reduce pesticide use while protecting crop yield and increasing crop production.

   Predicate Logic / First-Order Logic:

   $\forall x\ (\text{LeafSpot}(x) \rightarrow \text{ApplyPesticide}(x))$

   $\forall x\ (\text{HumidityHigh}(x) \land \text{TemperatureHigh}(x) \rightarrow \text{FungalRemoval}(x))$

5. Smart Resource Optimization

Ensuring efficient use of machinery, materials, and energy based on real-time sensor data and predictive AI models.

- *IF material_wastage_rate = increasing OR remaining_material_stock = low THEN reorder_materials = needed*
- *IF predicted_material_usage = high AND supplier_delivery_delay = long THEN early_procurement = required*

Explanation:

The system helps to minimize wastage and ensures materials are available when needed. If waste goes up or stock levels are too low, it suggests reordering to prevent being out of stock altogether. The system also forecasts material requirements. If it is expected to see a lot of usage, and the deliveries take a long time, it recommends ordering ahead so there are no delays.

Predicate Logic / First-Order Logic:

$\forall x\ (\text{highWastage}(x) \lor \text{lowStock}(x) \rightarrow \text{reorder}(x))$

$\forall x\ \forall y\ (\text{highUsagePrediction}(x) \land \text{longDelivery}(y) \rightarrow \text{earlyProcurement}(x, y))$

The first predicate indicates that if a material has high wastage or low stock, the system must reorder it. The second predicate shows that if future demand is high and delivery time is long, early procurement is required.